

Matching Division Stories, Drawings, and Equations

Grades 2–3 Operations and Algebraic Thinking

Matching division stories to equations, reading equal-group and array drawings, choosing the story an equation tells, and checking with multiplication fact families

The one idea behind every problem. A division sentence always says the same three things: **how many in all**, **how many groups**, and **how many in each group**. A story, a drawing, and an equation are three ways of telling that one idea. Your job is to move between them.

$\text{total in all} \div \text{number of groups} = \text{how many in each group}$
$\text{total in all} \div \text{how many in each group} = \text{number of groups}$

Reference card. No numbers are filled in — use the numbers given in each problem.

Directions.

- Every problem asks you to show the match, not just the answer: write the division equation, or draw the groups, or say which story fits.
- You may draw dots, circles, or an array in the work space — pictures are part of the work here.
- Write your final answer on the answer line.

1. Maya has 12 stickers. She shares them equally with 3 friends at her table, so that each friend gets the same number.

Write the division equation this story tells, then answer it: how many stickers does each friend get?

Answer: _____

2. The picture below shows 20 counters that have already been sorted into 4 equal groups.



Write the division equation that matches the picture, then answer it: how many counters are in each group?

Answer: _____

3. Here is a multiplication sentence with a missing number:

$$3 \times \square = 18$$

- (a) What number belongs in the box?
- (b) Write the division sentence that belongs to the same fact family and asks for that same missing number.

Answer: _____

- 4.** A teacher has 24 crayons. She packs them into boxes, and each box holds 6 crayons. This story does *not* tell you the number of groups — it tells you how many go in each group. Write the division equation the story tells, then answer it: how many boxes does she fill?

Answer: _____

- 5.** Look at this equation:

$$15 \div 5 = \square$$

Which story below matches the equation? Circle A or B, tell in one sentence how you know, then answer the equation.

- A.** There are 15 shells. They are put into 5 equal piles.
B. There are 15 shells in each of 5 buckets.

Answer: _____

6. The picture below shows 18 counters sorted into 3 equal groups.



Write the division equation that matches the picture, then answer it. Then write the multiplication equation that matches the same picture.

Answer: _____

7. A fact family uses the same three numbers over and over.

(a) Fill in the missing number: $7 \times \square = 28$

(b) Write all four sentences of this fact family — two multiplication sentences and two division sentences.

Put the missing number from part (a) on the answer line.

Answer: _____

8. Read carefully — one of these numbers is the total and one is the size of each group. A farm stand has 32 apples. The apples are put into baskets with 8 apples in each basket. Write the division equation this story tells, then answer it: how many baskets are used?

Answer: _____

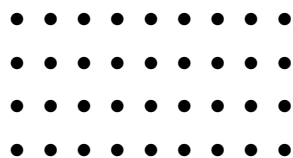
9. Look at this equation:

$$42 \div 6 = \square$$

- (a) Only one of these stories matches the equation. Circle it.
- A. A baker makes 42 muffins and puts 6 muffins in each tray.
- B. A baker makes 6 trays of muffins with 42 muffins on each tray.
- (b) Explain in one sentence why the other story does not match.
- (c) Answer the equation and write it on the answer line.

Answer: _____

10. The array below shows 36 counters arranged in 4 equal rows.



Write the division equation that matches the array, then answer it: how many counters are in each row? Be careful — the question asks about a row, not about the number of rows.

Answer: _____

11. This time the missing number is the one you divide *by*:

$$45 \div \square = 9$$

- (a) What number belongs in the box?
(b) Write a short story that this equation could tell, using your answer from part (a).
Put the missing number from part (a) on the answer line.

Answer: _____

12. Challenge. A classroom has 5 packs of pencils, and every pack holds 6 pencils. All of the pencils are shared equally among 3 tables.

- (a) What do you have to find *before* you can divide? Write that step as an equation.
- (b) Now write the division equation and answer it: how many pencils does each table get?
- (c) Draw a picture that shows your division. Label what the groups are.

Answer: _____